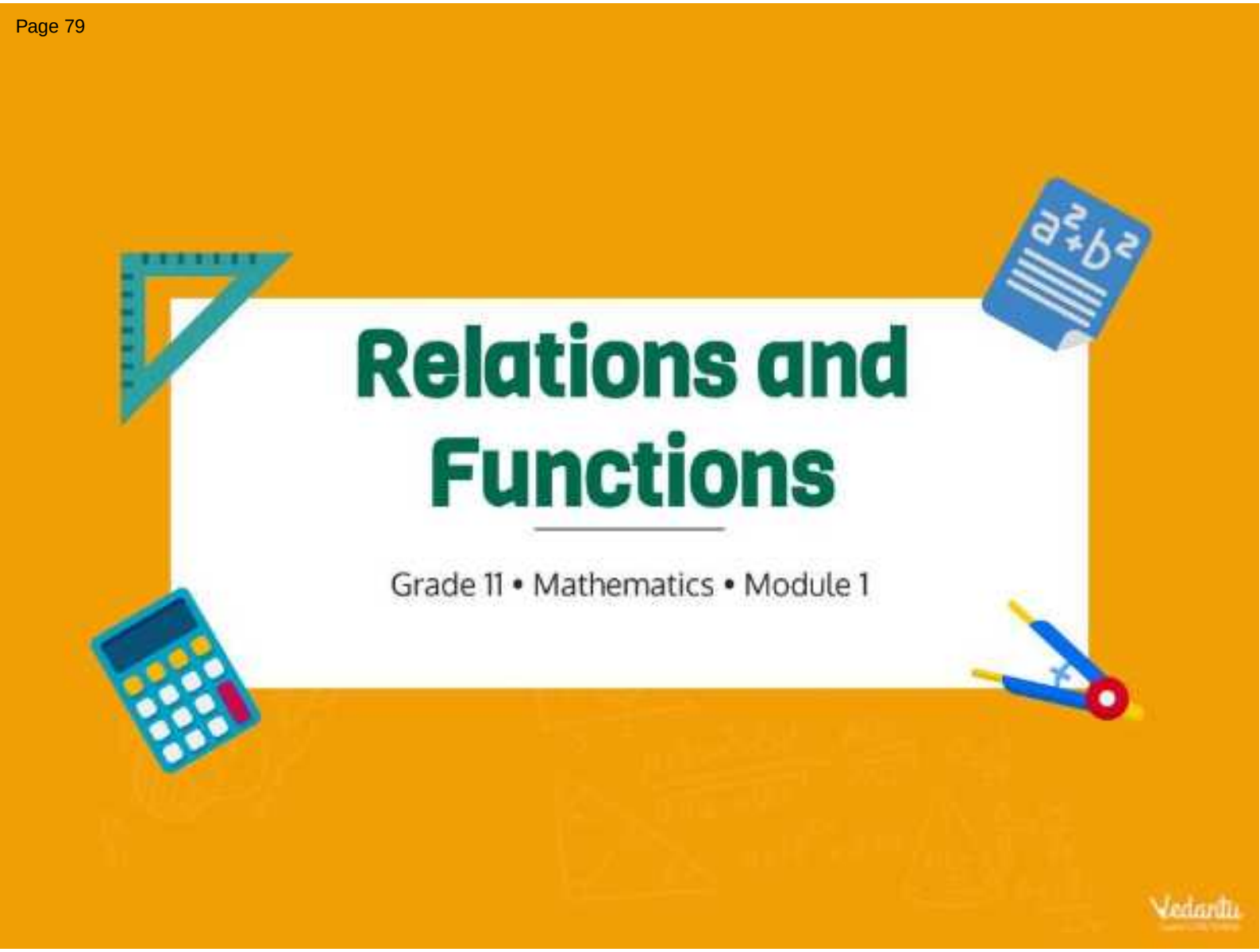

Relations and Functions
Grade 11 • Mathematics • Module 1



CAN YOU SEE & HEAR ME ?

Yes

No

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ORDERED PAIRS

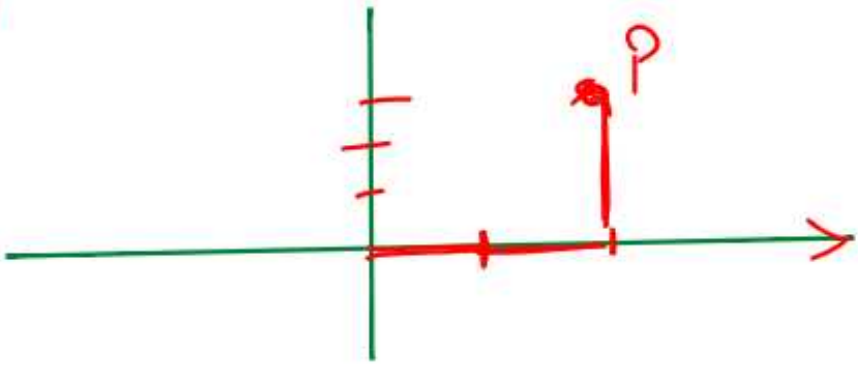
(x, y)

(y, x)

Coordinate of a point in 2D plane

x, y

$(2, 3)$



1.1 Cartesian Product of two Sets

$(A \times B)$

(x, y)

$A = \{1, 2, 3\}$

$B = \{a, b\}$

$n(A \times B) = n(B \times A) = n(A) \times n(B)$

$A \times B = \{(1, a), (1, b), (2, a), (2, b), (3, a), (3, b)\}$

$B \times A = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}$

$A \times B \neq B \times A$

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1.1 Cartesian Product of two Sets

Let A and B be two non-empty sets.

The Cartesian product of the sets A and B is defined as

$$A \times B = \{(a,b) : a \in A \text{ and } b \in B\}$$

Ex If $A = \{1,2\}$ and $B = \{a,b,c\}$, then

$A \times B =$

Ex If $A = \{1,2\}$, then

$A \times A =$

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1.1 Cartesian Product of two Sets

Let A and B be two non-empty sets.

The Cartesian product of the sets A and B is defined as

$$A \times B = \{(a,b) : a \in A \text{ and } b \in B\}$$

Ex If $A = \{1,2\}$ and $B = \{a,b,c\}$, then

$$A \times B = \{(1,a),(1,b),(1,c),(2,a),(2,b),(2,c)\}$$

Ex If $A = \{1,2\}$, then

$$A \times A = \{(1,1),(1,2),(2,1),(2,2)\}$$

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1.1 Cartesian Product of two Sets

Let A and B be two non-empty sets.

The Cartesian product of the sets A and B is defined as

$$A \times B = \{(a,b) : a \in A \text{ and } b \in B\}$$

Result

If $n(A) = p$ and $n(B) = q$, then $n(A \times B) = pq$

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Quiz.



Quiz.

Chat Off

Q. If $n(A)=3$ and $n(B)=2$ then number of subsets of $A\times B$ is

(A) 6 (B) 2^6 (C) 2^5 (D) None

$n(A\times B) = 3\times 2 = 6$

Total no. of subset $= 2^{n(A\times B)} = 2^6$

Quiz.

Chat On

Q. If $n(A)=3$ and $n(B)=2$ then number of subsets of $A\times B$ is

A 6

B 2^6

C 2^5

D None



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Quiz.

Q. If $n(A)=3$ and $n(B)=2$ then number of subsets of $A\times B$ is

Solution

Number of subsets of $A\times B = 2^{n(A)\times n(B)}$

$= 2^{3\times 2} = 2^6$



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1 Cartesian Product of Sets

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2. Relations

$R : A \to B$

Any subset of $A \times B$ = Relation

$A = \{1, 2, 3\}$

$B = \{a, b\}$

$A \times B = \{(1,a), (1,b), (2,a), (2,b), (3,a), (3,b)\}$

$R_1 = \{\}$ — (Empty Rel^n)

$R_2 = \{(2,a)\}$

$R_3 = \{(3,a), (3,b)\}$

$R_4 = \{(1,a), (1,b), (2,a), (2,b), (3,a), (3,b)\}$ — Universal Rel^n

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$R : A \to B$

Total no. of Relations = Total no of subsets of $A \times B$

$= 2^{n(A \times B)}$

$= 2^{n(A) \times n(B)}$

2. Relations

Definition

Let A and B be two non-empty sets.

Any subset of $A \times B$ is called a relation from A to B.

2. Relations

Definition

Let A and B be two non-empty sets.

Any subset of $A \times B$ is called a relation from A to B.

Ex Let $A = \{1, 2, 3\}$ and $B = \{5, 6\}$, then

$$A \times B = \{(1,5), (1,6), (2,5), (2,6), (3,5), (3,6)\}$$

Now, consider $R_1 = \{(1,5), (2,5), (3,6)$ and $R_2 = (2,6)\}$

Clearly R_1 and R_2, being subsets of $A \times B$, are relations from A to B.

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Let's Solve



Let's Solve

Ex. Find number of relations from $A=\{2,3,4,5\}$ to $B=\{a,b,c\}$.

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Let's Solve

Ex. Find number of relations from $A=\{2,3,4,5\}$ to $B=\{a,b,c\}$.

Solution

$n(A)=4$

$n(B)=3$

Number of relations $=2^{n(A)n(B)}=2^{4\times3}=2^{12}=4096$

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2. Relations

Result

Number of relations from A to $B = 2^{n(A)\times n(B)}$

Quiz.



Quiz.

Chat Off

Q. If $A=\{1,2,3\}$ and $B=\{5,6\}$, then the number of possible relations from A to B is _____.

A 6 **B** 5 **C** 2^6 **D** 2^5

$R: A \to B$

$n(A \times B) = 3 \times 2 = 6$

$\therefore$ no. of subset $= 2^6 =$ no. of Relation



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Quiz.

Chat On

Q. If $A = \{1,2,3\}$ and $B = \{5,6\}$, then the number of possible relations from A to B is _____.

A 6

B 5

C 2^6

D 2^5

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Quiz.

Q. If $A=\{1,2,3\}$ and $B=\{5,6\}$, then the number of possible relations from A to B is _____.

Solution

$A \quad \{1,2,3\}, n(A) \quad 3$

$B=\{5,6\}, n(B)=2$

Number of relation from A to $B = 2^{n(A)\times n(B)}$

$= 2^{3\times 2} = 2^6$



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2. Relations

Description of Relations

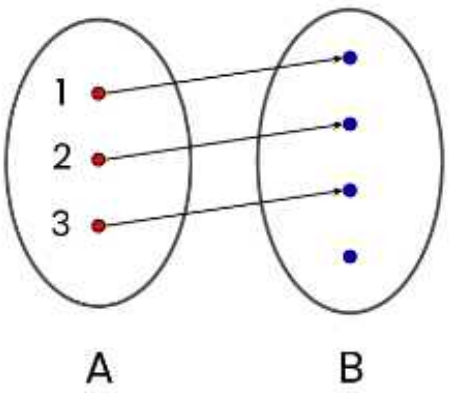
2. Relations

Description of Relations

There are three ways to describe a relation.

Let's do it by examples.

Let $A = \{1,2,3\}$ and $B = \{1,3,5,7\}$



(1) $R = \{(1,1),(2,3),(3,5)\}$

(or)

(2) $R = \{(a,b) : b = 2a-1, a \in A \text{ and } b \in B\}$

(or)

(3) $a\,R\,b \Leftrightarrow b = 2a-1; a \in A \text{ and } b \in B$

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2.1 Domain and Range of a Relation

$R : A \to B$

$A = \{1, 2, 3, 4, 5\}$

$B = \{3, 6, 9, 12, 15\}$

$R = \{(1, 6), (2, 9), (4, 12), (3, 15)\}$

Domain $= \{1, 2, 4, 3\}$

Co-domain $= \{3, 6, 9, 12, 15\}$

Range $= \{6, 9, 12, 15\}$

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2.1 Domain and Range of a Relation

Domain of $R = \{a : (a,b) \in R\}$

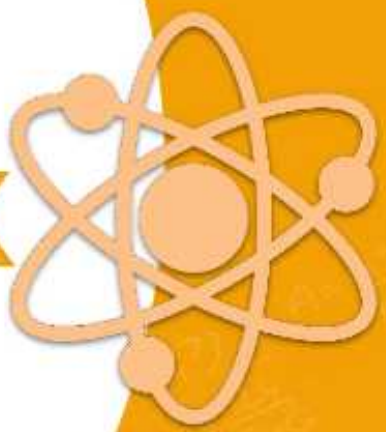
Range of $R = \{b : (a,b) \in R\}$

If R is a relation from A to B, then B is called co-domain.

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Exercise = 1 Solve (Assignment 1)

Feedback



Thanks.

